

GRE Probability Basics Cheat Sheet

1. Definition of Probability

The probability of an event A is given by:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Example: Rolling a fair six-sided die. Event A : Getting a 4

$$P(A) = \frac{1}{6}$$

2. Complementary Events

The complement of event A , denoted A' or $\overline{A}$, means “ A does not occur.”

$$P(A') = 1 - P(A)$$

Example: Probability of not rolling a 4 on a die:

$$P(A') = 1 - \frac{1}{6} = \frac{5}{6}$$

3. Addition Rule of Probability

For any two events A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Special Cases:

- If A and B are **mutually exclusive** (cannot occur together):

$$P(A \cap B) = 0 \quad \Rightarrow \quad P(A \cup B) = P(A) + P(B)$$

Example: Drawing one card from a deck:

$$P(\text{Ace or King}) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}$$

4. Multiplication Rule of Probability

For two events A and B :

$$P(A \cap B) = P(A) \times P(B|A)$$

If A and B are independent:

$$P(A \cap B) = P(A) \times P(B)$$

Example: Tossing two coins. Event A : First coin = Head Event B : Second coin = Head

$$P(A \cap B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

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5. Conditional Probability

The probability of event A given that B has occurred:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Example: From a deck of 52 cards, one card is drawn. Given that it's a face card, what's the probability it's a King?

$$P(\text{King} \mid \text{Face card}) = \frac{4/52}{12/52} = \frac{1}{3}$$

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6. Independent vs. Dependent Events

- **Independent:** Outcome of one event does not affect the other. $P(A \cap B) = P(A) \times P(B)$
 - **Dependent:** Outcome of one event affects the probability of the other. $P(A \cap B) = P(A) \times P(B|A)$
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7. Expected Value (Mean of a Random Variable)

If X is a discrete random variable taking values $x_1, x_2, \dots, x_n$ with probabilities $P(x_1), P(x_2), \dots, P(x_n)$, then:

$$E(X) = \sum_{i=1}^n x_i P(x_i)$$

Example: Toss a fair coin; win \$1 for Heads, lose \$1 for Tails.

$$E(X) = (1) \left(\frac{1}{2}\right) + (-1) \left(\frac{1}{2}\right) = 0$$

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8. Common Probability Scenarios

Scenario	Formula	Notes
Complementary Events	$P(A') = 1 - P(A)$	Opposite event
Mutually Exclusive	$P(A \cup B) = P(A) + P(B)$	No overlap
Independent Events	$P(A \cap B) = P(A)P(B)$	One doesn't affect other
Conditional Probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$	B occurred
At least one occurs	$1 - P(\text{none})$	Shortcut for "at least one" problems

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If A and B are independent:

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Example: Tossing two coins. Event A : First coin = Head Event B : Second coin = Head

$$P(A \cap B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

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$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

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Example: Toss a fair coin; win \$1 for Heads, lose \$1 for Tails.

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Summary

- Probability ranges between 0 and 1
- Total probability of all outcomes = 1
- Use **addition rule** for OR problems
- Use **multiplication rule** for AND problems
- Identify if events are independent or dependent
- Don't forget complements and “at least one” shortcuts